

Angular Velocity Calibration

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1. Course Content

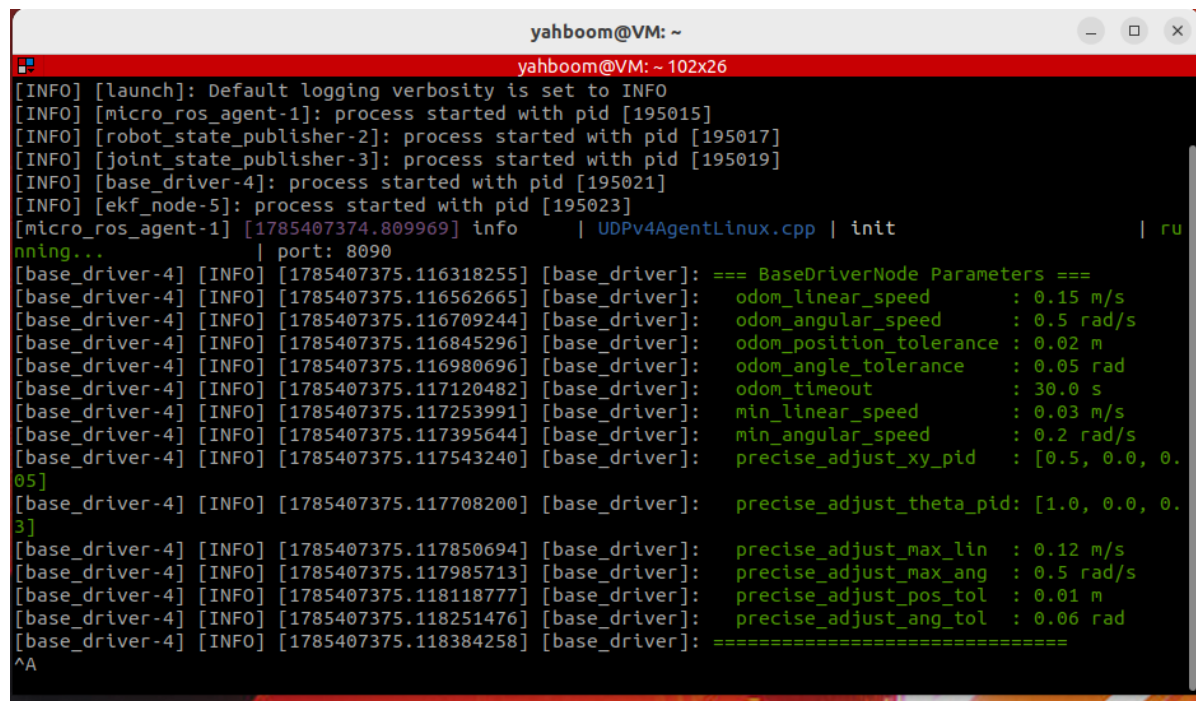
- Verify the rotation accuracy of the chassis wheel odometry (calculated from motor encoder data) and calibrate the angular velocity

[!IMPORTANT]

The car is calibrated at the factory, so no re-calibration is needed on first use. If you find that the odometry rotation error is large during subsequent use, refer to this tutorial for calibration.

2. Preparation

```
ros2 launch microros_v2_bringup car_base.launch.py use_camera:=false
```



```
yahboom@VM: ~
yahboom@VM: ~ 102x26
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [micro_ros_agent-1]: process started with pid [195015]
[INFO] [robot_state_publisher-2]: process started with pid [195017]
[INFO] [joint_state_publisher-3]: process started with pid [195019]
[INFO] [base_driver-4]: process started with pid [195021]
[INFO] [ekf_node-5]: process started with pid [195023]
[micro_ros_agent-1] [1785407374.809969] info | UDPv4AgentLinux.cpp | init | ru
nning... | port: 8090
[base_driver-4] [INFO] [1785407375.116318255] [base_driver]: === BaseDriverNode Parameters ===
[base_driver-4] [INFO] [1785407375.116562665] [base_driver]: odom_linear_speed : 0.15 m/s
[base_driver-4] [INFO] [1785407375.116709244] [base_driver]: odom_angular_speed : 0.5 rad/s
[base_driver-4] [INFO] [1785407375.116845296] [base_driver]: odom_position_tolerance : 0.02 m
[base_driver-4] [INFO] [1785407375.116980696] [base_driver]: odom_angle_tolerance : 0.05 rad
[base_driver-4] [INFO] [1785407375.117120482] [base_driver]: odom_timeout : 30.0 s
[base_driver-4] [INFO] [1785407375.117253991] [base_driver]: min_linear_speed : 0.03 m/s
[base_driver-4] [INFO] [1785407375.117395644] [base_driver]: min_angular_speed : 0.2 rad/s
[base_driver-4] [INFO] [1785407375.117543240] [base_driver]: precise_adjust_xy_pid : [0.5, 0.0, 0.
05]
[base_driver-4] [INFO] [1785407375.117708200] [base_driver]: precise_adjust_theta_pid: [1.0, 0.0, 0.
3]
[base_driver-4] [INFO] [1785407375.117850694] [base_driver]: precise_adjust_max_lin : 0.12 m/s
[base_driver-4] [INFO] [1785407375.117985713] [base_driver]: precise_adjust_max_ang : 0.5 rad/s
[base_driver-4] [INFO] [1785407375.118118777] [base_driver]: precise_adjust_pos_tol : 0.01 m
[base_driver-4] [INFO] [1785407375.118251476] [base_driver]: precise_adjust_ang_tol : 0.06 rad
[base_driver-4] [INFO] [1785407375.118384258] [base_driver]: =====
^A
```

3. Calibration Demonstration

Note: If the odometry data is wrong, you can press the reset button to restart the car.

- Run the command in the terminal

```
ros2 run common_demo calibrate_angular_z --ros-args -p test_angle_deg:=360.0
```

- The tested angle test_angle_deg here is 360 degrees. After the car stops moving, measure the actual rotation angle and enter the measured actual rotation angle into the terminal; the terminal will automatically calculate the calibration coefficient.

```
=====
Please enter actual rotation angle (degrees): 360
[INFO] [1785412340.750292582] [calibrate_angular_z]: =====
[INFO] [1785412340.750496594] [calibrate_angular_z]: Calibration Results
[INFO] [1785412340.750744268] [calibrate_angular_z]: =====
[INFO] [1785412340.751011943] [calibrate_angular_z]: Odometry reading: 359.07 degrees (6.2669 radians)
[INFO] [1785412340.751182873] [calibrate_angular_z]: Actual angle: 360.00 degrees (6.2832 radians)
[INFO] [1785412340.751332704] [calibrate_angular_z]: Correction factor: 1.0026
[INFO] [1785412340.751551151] [calibrate_angular_z]: =====
yahboom@VM:~$
```

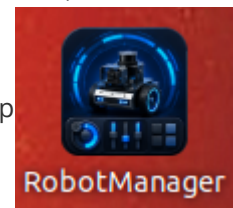
[!TIP]

test_angle_deg:=360° is a demonstration example; you can adjust the tested angle as needed.

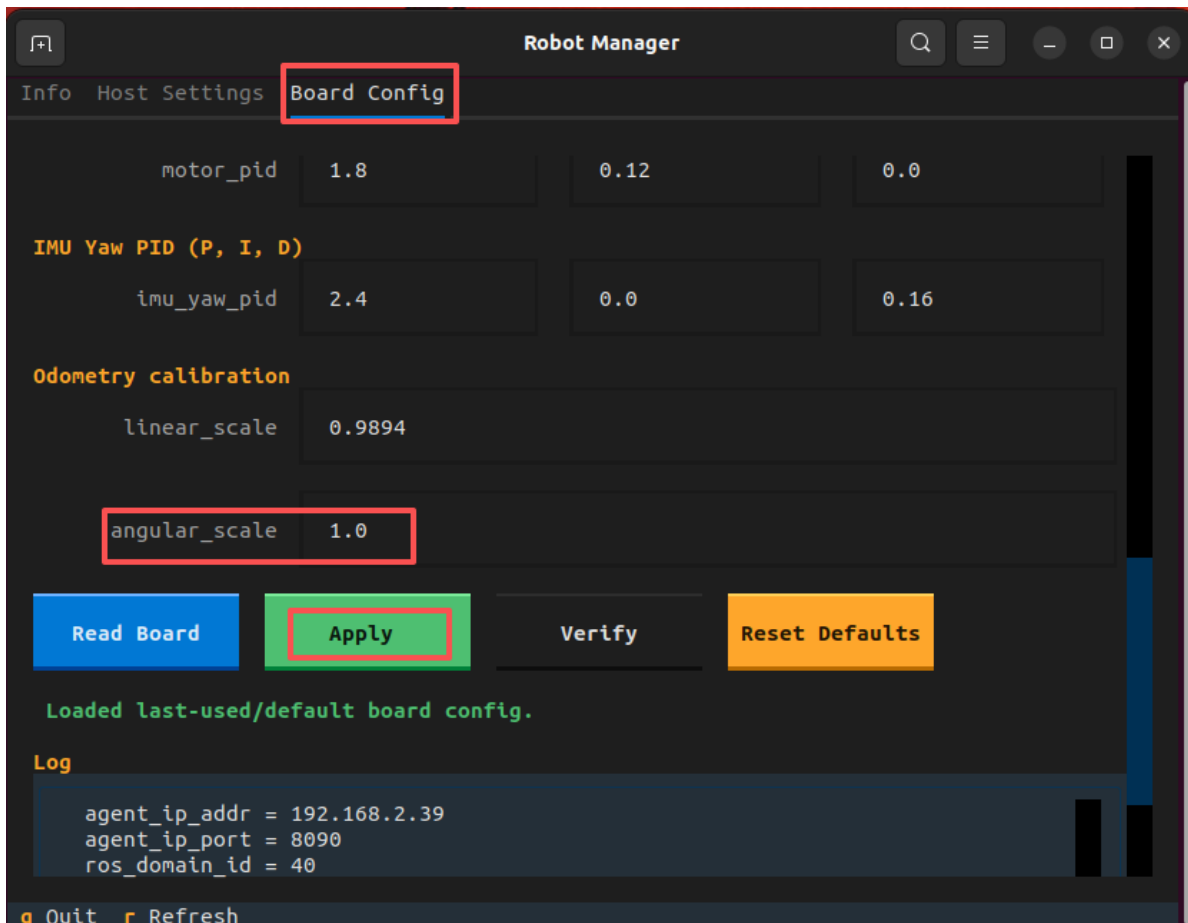
4. Write the Calibration Coefficient to the MicroRos Control Board

- Close the microros_v2_bringup and calibrate_linear terminals started earlier, and double-

click the RobotManager application icon on the virtual machine desktop



- On the control board page, enter the calibration coefficient calculated in the previous step into the angular velocity coefficient, and click Apply Configuration.



5. Verification Test

- Repeat the test again `ros2 run common_demo calibrate_angular_z --ros-args -p test_angle_deg:=360.0`; the rotation error will now be smaller than before calibration.